docs_sect_4_2_mars_MOC

Processed on: 2025-10-19 11:20:03

ASP version: 3.6.0-alpha

asp_plot version: 1.9.0

DEM Summary

Property	Value
DEM File	output-DEM.tif
Dimensions (px)	1538 x 5365
GSD (m)	3.41
CRS	EPSG:None
Nodata (%)	46.0
Elevation Range (m)	-10531.1 to -9967.6

Input Scenes

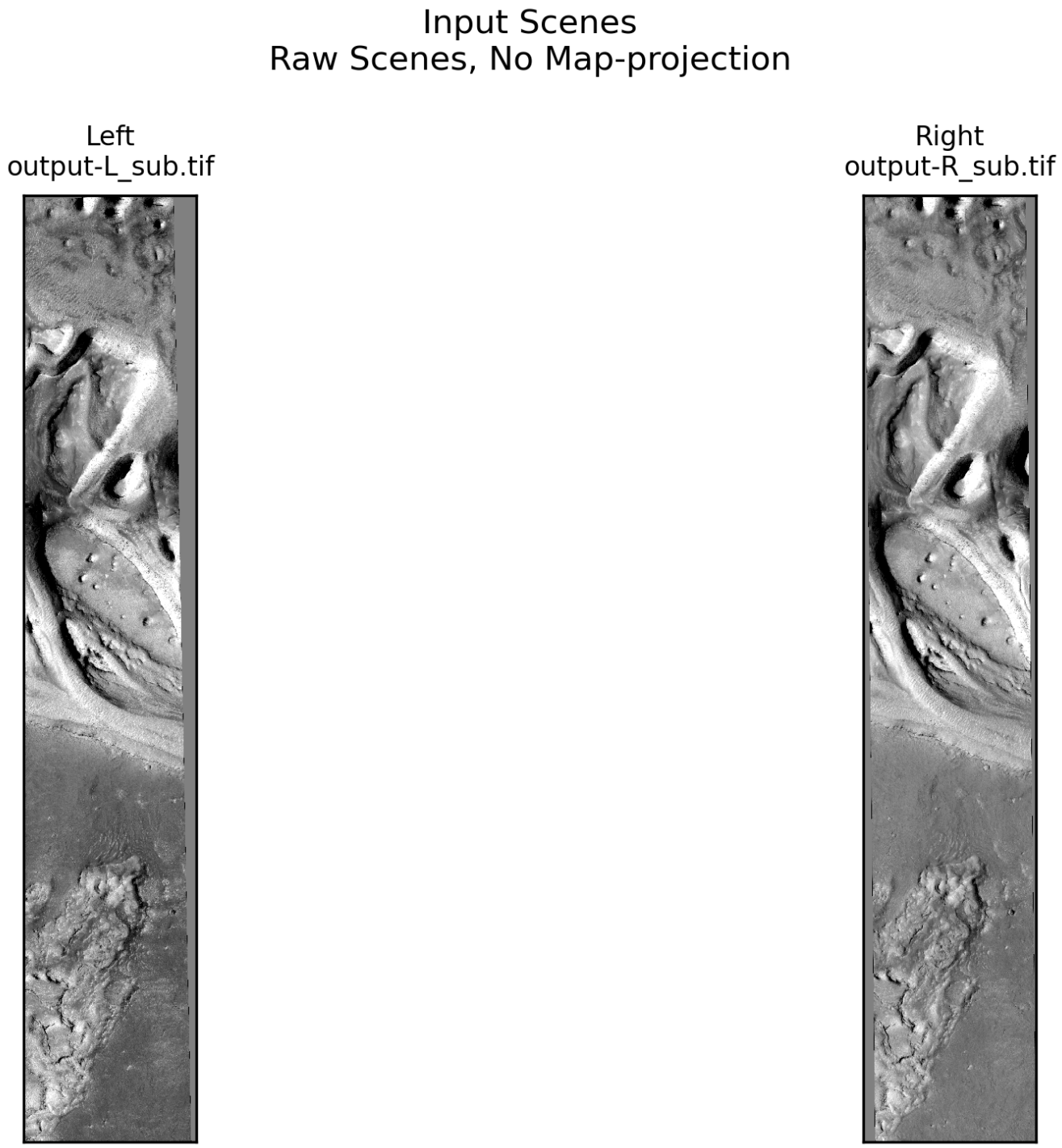

Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., affineepipolar), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes require no pre-alignment and are displayed in their map-projected orientation.

Match Points

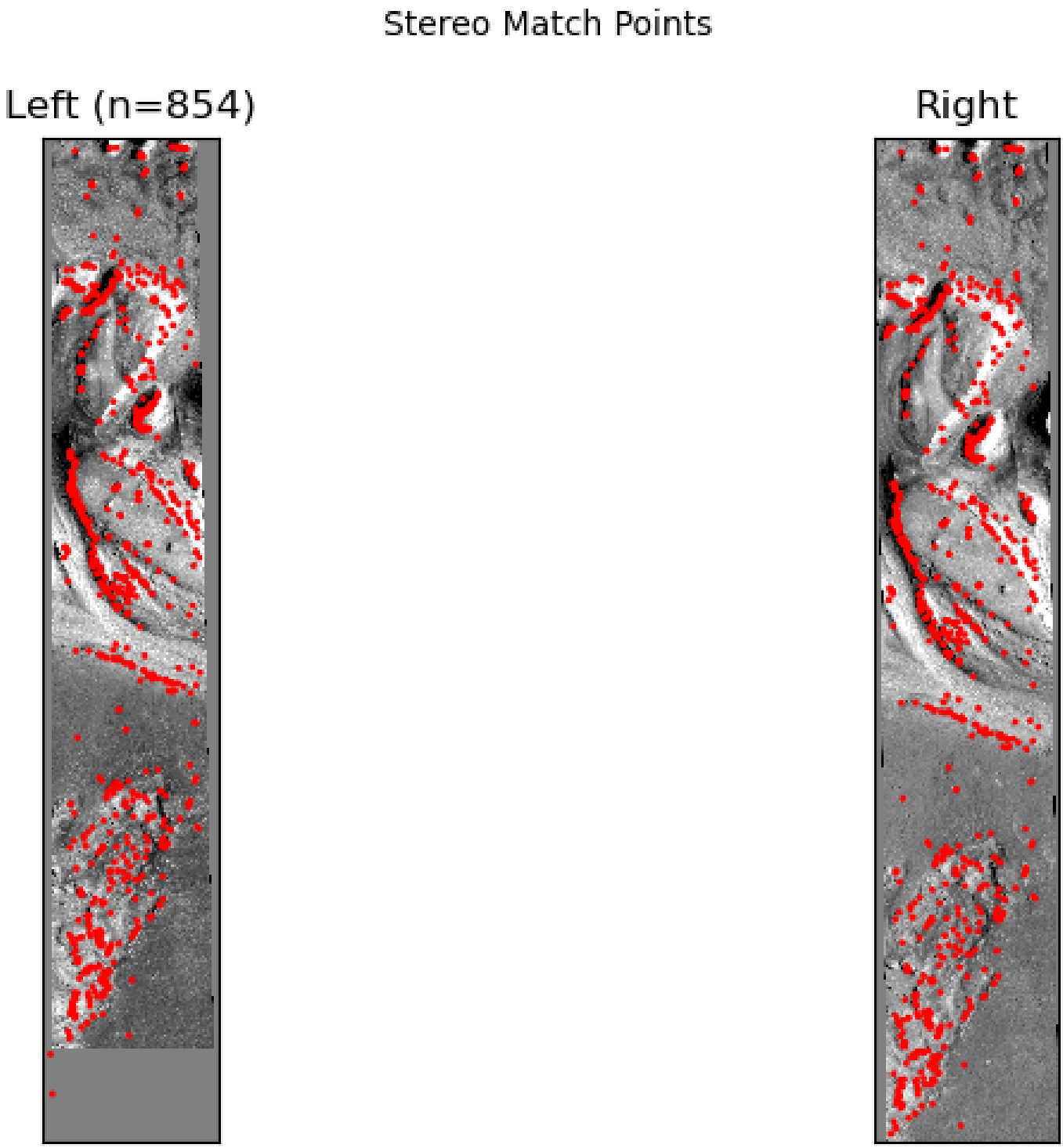

Figure 2: Interest point matches between left and right images identified during stereo correlation.

Detailed Hillshade

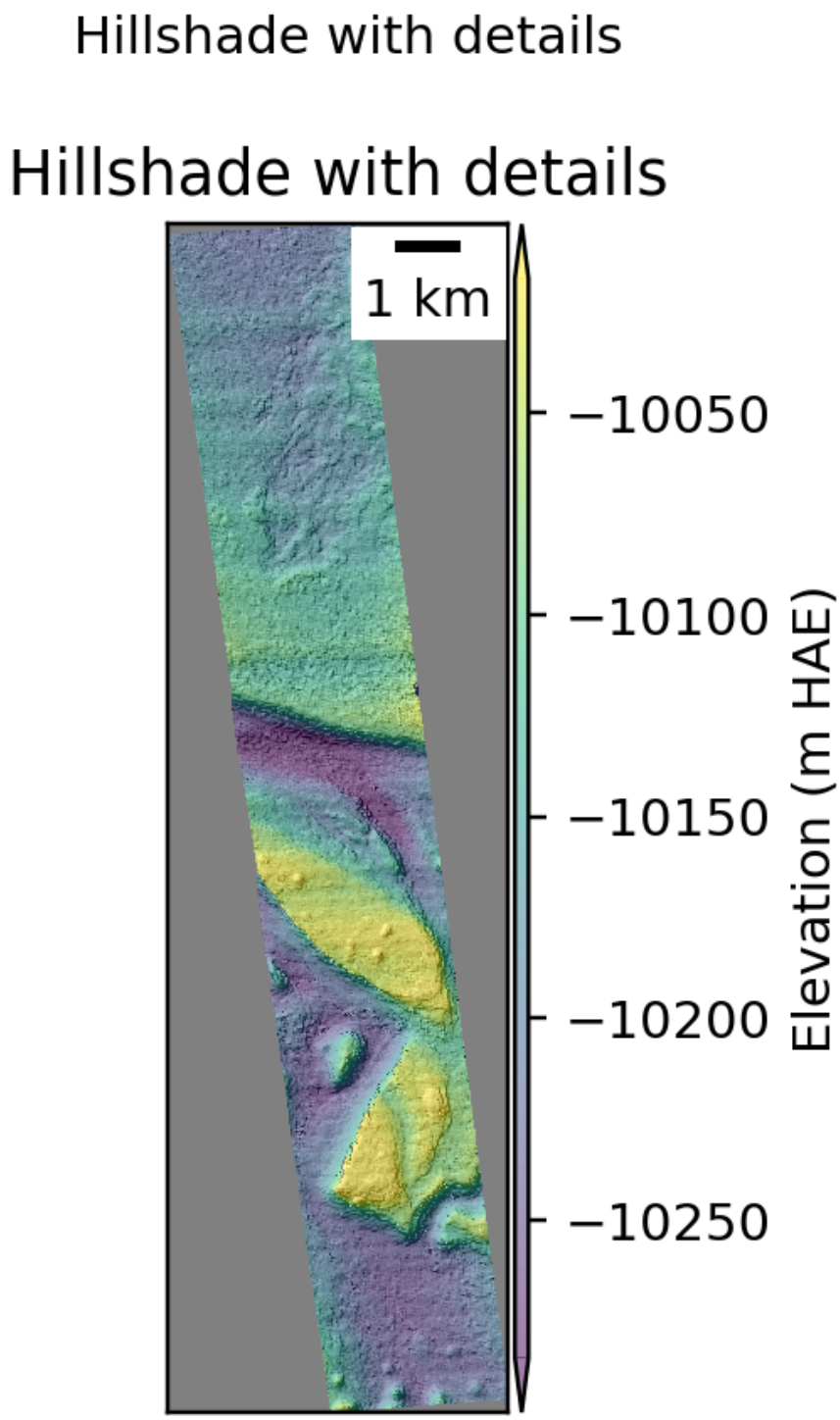

Figure 3: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.

DEM Results

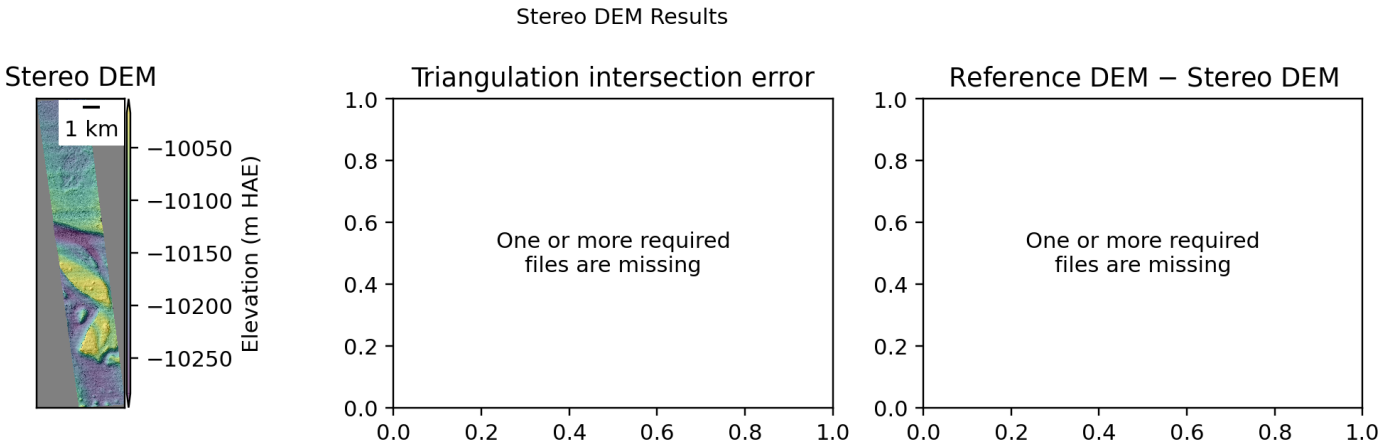

Figure 4: Output DEM with intersection error map and difference relative to the reference DEM used in processing.

Disparity

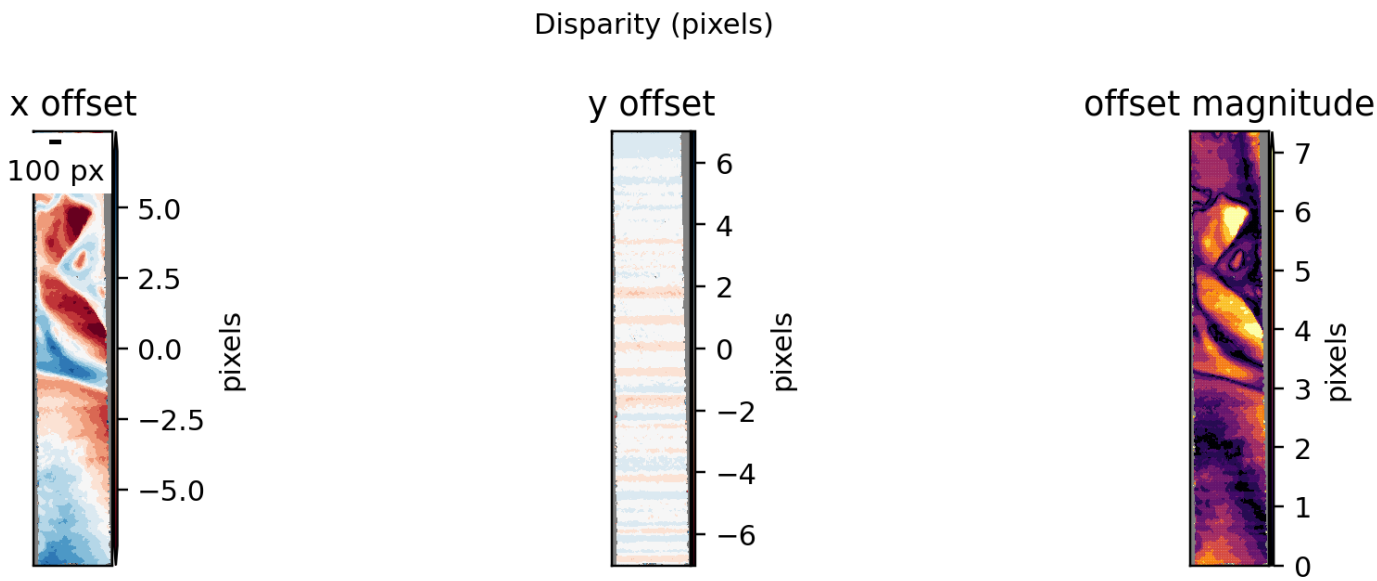

Figure 5: Horizontal and vertical disparity maps in pixels with quiver overlay.

MOLA Altimetry Map

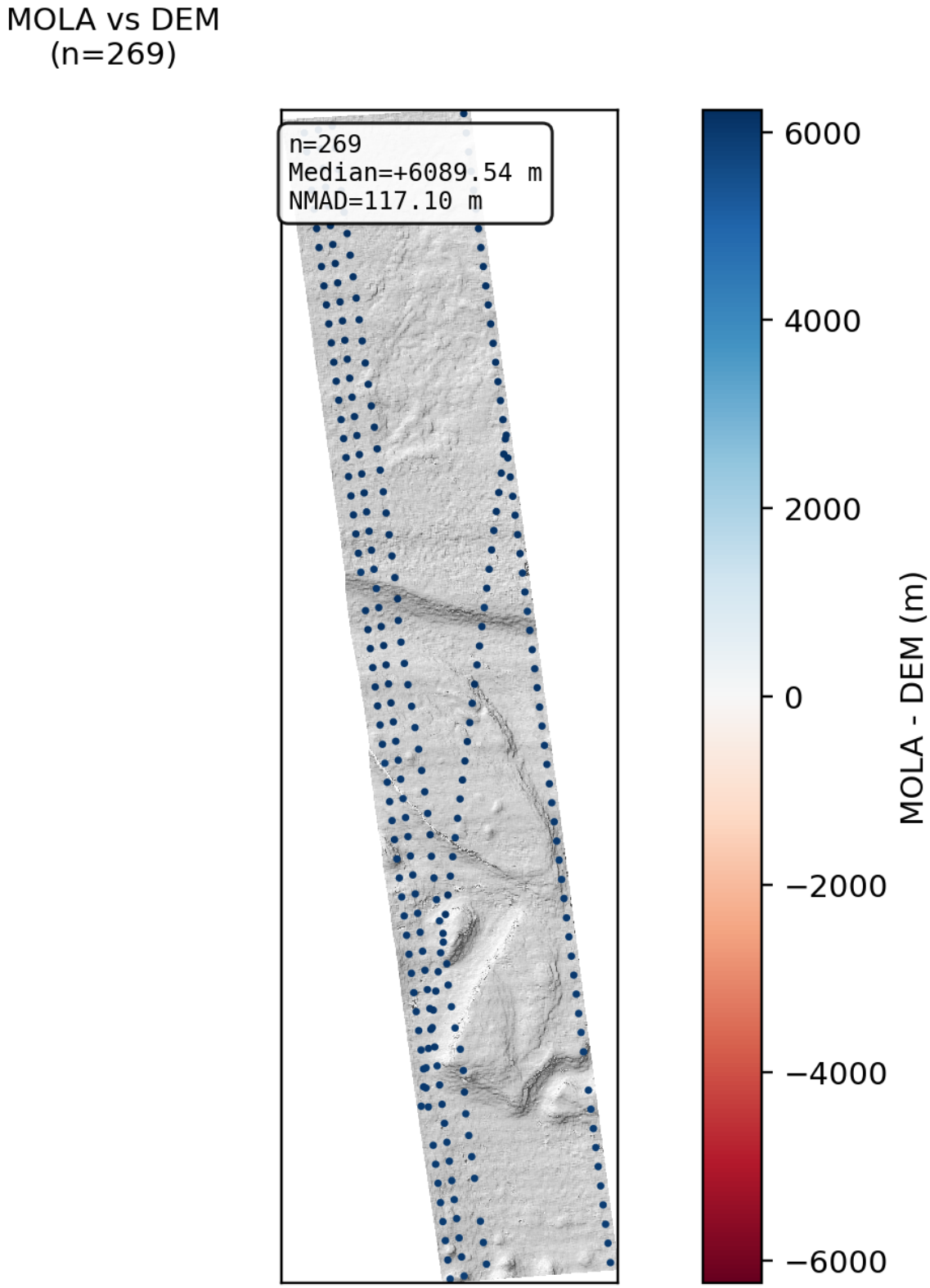

Figure 6: MOLA elevation differences vs. ASP DEM.

MOLA Altimetry Histogram

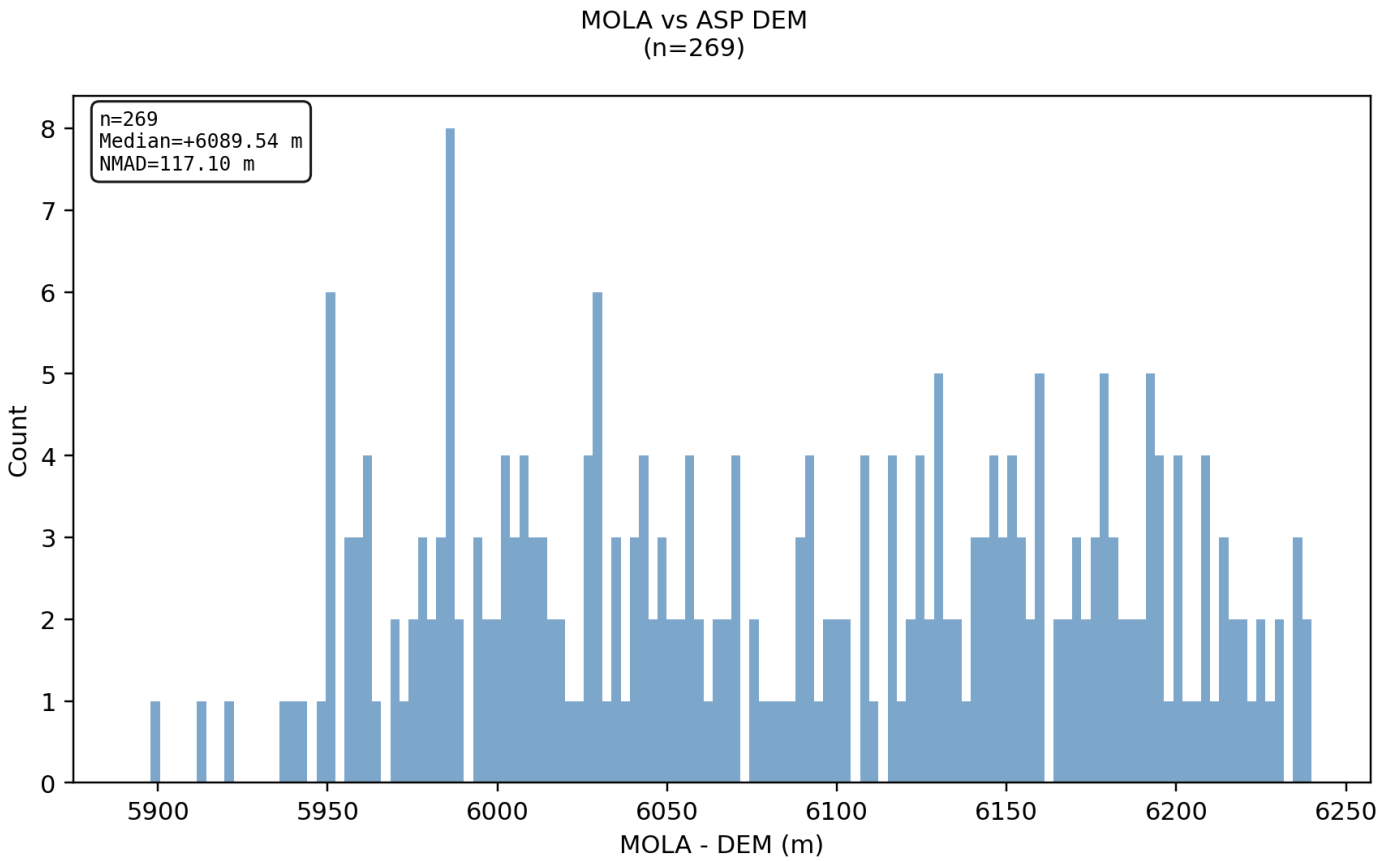

Figure 7: Distribution of elevation differences between MOLA and ASP DEM.

Processing Parameters

Runtime Summary

Step	Runtime
Bundle Adjust	N/A
Stereo	0 hours and 2 minutes
point2dem	0 hours and 0 minutes

Bundle Adjust Command:

```
Bundle adjustment not run
```

Stereo Command:

```
stereo --alignment-method affineepipolar E0201461.cub M0100115.cub results/output --corr-seed-mode 1 --stereo-file
stereo.default.example --sgm-collar-size 0 --compute-point-cloud-center-only --threads 8
```

point2dem Command:

```
point2dem --auto-proj-center results/output-PC.tif
```

Report Generation Command:

```
asp_plot --directory /Users/ben/Dropbox/UW_Shean/asp_plot-test_data/docs_examples/docs_sect_4_2_mars_MOC/
--stereo_directory results/ --add_basemap False --altimetry_csv /Users/ben/Dropbox/UW_Shean/asp_plot-
test_data/docs_examples/docs_sect_4_2_mars_MOC//438115_mola/MolaPEDR_34N34N_142E142E_20260319T204814563_topo_csv.csv
--plot_geometry False --subset_km 0.1
```